AM 121 EO1 Group 16

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Task 1: The Mathematical Formulation

1(a): Base Mathematical Model With Set Limits

Goal

Determine the intensity of each radiation beam to maximize dosage over tumor areas while minimizing dosage over critical areas, subject to upper and lower limits on radiation exposure for the critical and tumor areas respectively.

Sets

$b \in B:$ the discrete set of beams used during the radiotherapy treatment

$p \in P:$ the discrete set of identifiable pixels in the provided 2D image of the tumorous and critical areas

$t \in T:$ the set of pixels, t, in the tumorous area included in P, the total set of identifiable pixels p, such that $T \subseteq P$

$c \in C:$ the set of pixels, t, in the critical area included in P, the total set of identifiable pixels p, such that $C \subseteq P$

Parameters

$a_{p,b}:$ the relative intensity of beam $b \in B$ directed at pixel $p \in P,$ normalized between 0 and 1 inclusively

$R_T:$ the pre-set minimum radiation limit required uniformly for the pixels in the tumor area, $\forall t \in T$

$R_C:$ the pre-set maximum radiation limit allowed uniformly for pixels in the critical area, $\forall c \in C$

Decision Variables

$x_b:$ the intensity of radiation assigned by to the beam $b \in B$ for the radiotherapy treatment

Objective Function (Final Minimize Beam Intensity Version)

$$\min\ Z = \sum_{b \in B} x_b \qquad \text{minimizes the total radiation beam intensity of all the } b \in B$$

This objective function can be understood as minimizing the total amount of radiation used during the radiotherapy treatment while still satisfying the required constraints. This is ideal as the oncologist will not want to subject the patient to more radiation than is required for the tumor treatment. However, this assumption would mean that that higher radiation intensity that puts directs the excess radiation only toward the tumor region while not impacting the total radiation received by the critical area would not be

considered an optimal improvement, which is not necessarily realistic as a higher radiation dosage received by tumor cells would be a more effective treatment.

Objective Function (Original Maximize Tumor Radiation and Minimize Critical Radiation)

$$\max\ Z = \sum_{p\in T}\sum_{b\in B} a_{p,b}x_b\ -\ \sum_{p\in C}\sum_{b\in B} a_{p,b}x_b$$
$$\max\ Z = \sum_{b\in B} x_b(\sum_{p\in T} a_{p,b}\ -\ \sum_{p\in C} a_{p,b})$$

This objective function can be understood as maximizing the total amount of radiation received by the tumor area while minimizing the amount of radiation received by the critical area. This objective function works by reflecting our primary goal of maximizing the amount of radiation directed at the tumor area in order to treat it, while also penalizing against higher values of radiation directed at the critical region in the process of targeting the tumorous region.

Constraints

$$\sum_{b\in B} a_{t,b}x_b \geq R_T, \qquad \forall\, t\in T \tag{1}$$
$$\sum_{b\in B} a_{c,b}x_b \leq R_C, \qquad \forall\, c\in C, \tag{2}$$
$$x_b \geq 0, \qquad \forall\, b\in B. \tag{3}$$

Constraint (1) requires that the sum total radiation received by each tumor pixel from each beam is at least the minimum radiation dosage for tumor cell treatment, R_T.
Constraint (2) requires the sum total radiation received by each critical pixel from each beam to be less than the maximum radiation allowed for tumor cells, R_C.
Constraint (3) limits the radiation intensity level of each beam to be non-negative as a negativity intensity level is not possible for the problem

Model Summary

$$\max_x \quad \sum_{t\in T}\sum_{b\in B} a_{t,b}x_b - \sum_{c\in C}\sum_{b\in B} a_{c,b}x_b \qquad \textbf{or} \qquad \min Z = \sum_{b\in B} x_b$$
$$\text{s.t.} \quad \sum_{b\in B} a_{t,b}x_b \geq R_T, \qquad \forall\, t\in T \tag{1}$$
$$\sum_{b\in B} a_{c,b}x_b \leq R_C, \qquad \forall\, c\in C \tag{2}$$
$$x_b \geq 0, \qquad \forall\, b\in B \tag{3}$$

Task 1(b): Allowing Slight Violations for Feasibility

Goal

Modify the base model from Task 1(a) to allow for slight variation on the set radiation limits of the tumor and critical areas where it is necessary to ensure the feasibility of the model, which is the case only if the limits cannot be met.

Additional Decision Variables

δ_t : the value of the deficit between the total radiation received by tumor pixel $t \in T$ and its minimum limit value

δ_c : the value of the surplus between the total radiation received by critical pixel $c \in C$ and its maximum limit value

Modified Constraints

$\sum_{b \in B} a_{t,b} x_b + \delta_t \geq R_T, \quad \forall\, t \in T$ adds a artificial variable to allow a tumor pixel to receive less than its minimum radiation

$\sum_{b \in B} a_{c,b} x_b - \delta_c \leq R_C, \quad \forall\, c \in C$ adds a artificial variable to allow a critical pixel to receive more than its allowed radiation

$x_b,\ \delta_p^T,\ \delta_p^C \geq 0,$ requires that all variables be non-negative in value

Minimization Objective Function

$$\min_{x_b, \delta_t, \delta_c} \quad Z = \sum_{b \in B} x_b$$

By keeping the same objective function, we still minimize the total radiation intensity of the beams used during treatment. However, in order to check the feasibility of the problem with this objective function, we need to minimize the value of our artificial decision variables via the Simplex phase 1 method.

Simplex Phase 1: Minimizing Total Variation From Radiation Limits

$$\begin{aligned}
\min_{x, \delta_t, \delta_c} \quad & V = \sum_{t \in T} \delta_t + \sum_{c \in C} \delta_c \\
\text{s.t.} \quad & \sum_{b \in B} a_{t,b} x_b + \delta_t \geq R_T, && \forall t \in T \\
& \sum_{b \in B} a_{c,b} x_b - \delta_c \leq R_C, && \forall c \in C \\
& x_b \geq 0,\ \delta_t \geq 0,\ \delta_c \geq 0
\end{aligned}$$

Let $V^\star$ be the optimal solution for this phase representing the optimal total variation from the fixed radiation limits.

Simplex Phase 2: Minimizing Total Beam Intensity

$$
\begin{aligned}
\min_{x_b,\delta_t,\delta_c} \quad & Z = \sum_{b\in B} x_b \\
\text{s.t.} \quad & \sum_{b\in B} a_{t,b}x_b + \delta_t \geq R_T, && \forall t \in T \\
& \sum_{b\in B} a_{c,b}x_b - \delta_c \leq R_C, && \forall c \in C \\
& \sum_{t\in T} \delta_t + \sum_{c\in C} \delta_c \leq V^\star \\
& x_b,\ \delta_t,\ \delta_c \geq 0
\end{aligned}
$$

Simplex Phase 1 guarantees feasibility by finding the smallest necessary aggregate relaxation of the radiation limits, and Phase 2 computes the minimal-intensity beam configuration that respects that same total deviation. By splitting the problem up into two phases, we can solve for the minimum possible variation from the pre-set radiation limits imposed on the tumor and critical areas that will allow the Simplex Phase 2 model to be feasible. Then by solving the Simplex Phase 2 model with these optimal Phase 1 values for our added artificial variables representing the allowed variation from the set radiation limits, we are able to find the corresponding optimal solution to our objective function of minimizing total radiation intensity while meeting the required constraints for both the critical and tumor areas.

Task 1(c): Penalize Dose on Non-Critical Boundary Near Critical Tissue

Additional Sets and Parameters

$$
\begin{aligned}
i \in PR: &\quad \text{the set of pixel rows in the provided 2D photo} \\
j \in PC: &\quad \text{the set of pixel columns in the provided 2D photo} \\
h \in H = P \setminus (T \cup C): &\quad \text{the set of pixels, } h \in P\text{, that are non-tumor pixels, } h \notin T\text{, and non-critical, } h \notin C \\
n \in N(p) \subseteq P: &\quad \text{the set of pixels, } n \in P\text{, bordering pixel, } p \in P\text{, within by a set offset region} \\
e \in E_{h,N_{di,dj}} = \{\, p \in H : \exists\, e \in C \text{ with } e \in N(p) \,\}: &\ \text{the set of non-critical pixels on the border of the critical area} \\
\gamma > 0: &\ \text{the penalty weight for radiation delivery to pixels, } e \in E\text{, that are non-critical but border a critical area} \\
\lambda_T: &\ \text{The penalty weight applied to deviating from the imposed radiation limit on tumor cells} \\
\lambda_C: &\ \text{The penalty weight applied to deviating from the imposed radiation limit on critical cells}
\end{aligned}
$$

Neighborhood definition:
N_{d_i,d_j} = the neighborhood of pixels around the pixel $h \in H$ located at $(i,j) \in PC \times PR$, defined by the offset:

$$d_i, d_j \in \{-1, 0, 1\}, \qquad \forall d_i, d_j$$

Another way of stating this is to let N_{d_i,d_j} denote the neighborhood of pixels around pixel p in H, indexed at (i,j) with $d_i, d_j \in \{-1,0,1\}$ for all d_i, d_j.

Objective With Non-Critical Boundary Radiation Delivery Penalty

$$\min\ W = \lambda_T \sum_{t\in T} \delta_t - \lambda_C \sum_{c\in C} \delta_c - \gamma \sum_{e\in E}\sum_{b\in B} a_{e,b} x_b$$

(The new term discourages dose on non-critical pixels that border critical tissue; choose γ small so that its a small penalty)

Constraints (as in 1b).

$$\sum_{b\in B} a_{t,b} x_b + \delta_t \geq R_T, \qquad \forall\, p \in T, \tag{1}$$

$$\sum_{b\in B} a_{c,b} x_b - \delta_c \leq R_C, \qquad \forall\, p \in C, \tag{2}$$

$$x_b,\ \delta_p^T,\ \delta_p^C \geq 0. \tag{3}$$

Model summary.

$$\begin{aligned}
\min \quad & Z = \lambda_T \sum_{t\in T} \delta_t - \lambda_C \sum_{c\in C} \delta_c - \gamma \sum_{e\in E}\sum_{b\in B} a_{p,b} x_b \\
\text{s.t.} \quad & \sum_{b\in B} a_{t,b} x_t + \delta_t \geq R_T, && \forall t \in T, \\
& \sum_{b\in B} a_{c,b} x_b - \delta_c \leq R_C, && \forall c \in C, \\
& x_b,\ \delta_t,\ \delta_c \geq 0.
\end{aligned}$$

Task 2a: Running the Model on the Small and Actual Examples

Objective. Run the Task 1a model in AMPL on both datasets and report feasibility/results.
AMPL files used.

- `task2a.mod` — model ($a_{p,b}$, R_T, R_C, beam vars x_b, dose link, tumor/critical limits).
- `task2a_small.dat`, `task2a_actual.dat` — data for small vs. actual.
- `task2a_small.run`, `task2a_actual.run` — runs that solve and export outputs to `2a_small_beam.out`, `2a_small_intensities.out`, `2a_actual_beam.out`, `2a_actual_intensities.out`. We converted these files to CSVs forvisualization.

Small example — feasible.

```
CPLEX 22.1.1: optimal solution; objective 52.1
0 simplex iterations
```

Minimum total beam intensity = 52.1. All tumor pixels satisfy dose $\geq R_T = 10$ and all critical pixels satisfy dose $\leq R_C = 2$. The beam vector and dose matrix are saved in `2a_small_beam.out` and `2a_small_intensities.out`.

Actual example — infeasible (presolve contradiction). Running `task2a_actual.run` produces presolve messages that certify infeasibility before simplex:

```
presolve: constraint DoseDef[26,42] cannot hold:
  body <= 0 cannot be >= 0.277426; difference = -0.277426
presolve: constraint DoseDef[26,41] cannot hold:
  body <= 0 cannot be >= 0.311688; difference = -0.311688
presolve: constraint DoseDef[26,40] cannot hold:
  body <= 0 cannot be >= 0.182511; difference = -0.182511
presolve: constraint DoseDef[25,44] cannot hold:
  body <= 0 cannot be >= 0.270522; difference = -0.270522
presolve: constraint DoseDef[25,42] cannot hold:
  body <= 0 cannot be >= 0.275681; difference = -0.275681
... (414 similar lines suppressed)
```

What this means. We enforce the dose link

$$\text{dose}[i,j] \;=\; \sum_b a_{b,i,j}\, x_b.$$

For several tumor pixels (i,j) in the actual dataset, the coverage satisfies $\sum_b a_{b,i,j} = 0$ (no beam reaches that pixel), while the model requires $\text{dose}[i,j] \geq R_T > 0$. Combining these gives $0 \geq R_T$, a direct contradiction. AMPL's presolve detects this and stops, so the Task 1a model is **infeasible** on the actual instance; there is no beam vector that meets R_T everywhere in T while respecting physics and R_C.

Why this sets up Task 2b. Because some tumor pixels are unreachable with the provided beams, we relax the limits with nonnegative slacks. Phase 1 minimizes total violation $V^{=\sum_{t\in T}\delta_T[t]+\sum_{c\in C}\delta_C[c]}$ to find the smallest aggregate deviation needed for feasibility; Phase 2 then minimizes $\sum_b x_b$ subject to total violation $\leq V$. This yields the least-intensity plan consistent with the minimal relaxation.

Task 2b: Feasibility Relaxation and Two-Phase Solve

Objective. Allow per-pixel slacks so the model can solve when strict limits make the instance infeasible (actual dataset). Phase 1 minimizes total violation; Phase 2 minimizes total beam intensity subject to the Phase 1 violation budget.
AMPL files used.

- `task2b.mod` — model with pixel-level slacks and two objectives (`TotalViolation`, `TotalIntensity`).
- `task2b_small.run`, `task2b_actual.run` — two-phase runs that export beam vectors and dose matrices to `2b_small_beam.csv`, `2b_small_intensities.csv`, `2b_actual_beam.csv`, `2b_actual_intensities.csv`.

Small example — feasible (no relaxation needed). Phase 1 returns $V^{=0}$. Under the $V^{=0}$ cap, Phase 2 reproduces the same feasible plan as in 2a while minimizing total intensity. The optimal total beam intensity is **52.1** with 4/5 beams nonzero. Tumor constraints (dose $\geq R_T = 10$) and critical constraints (dose $\leq R_C = 2$) are satisfied, matching the 2a solution.

Actual example — feasible with allowable slacks. With slacks permitted, Phase 1 identifies a positive V, confirming that a small aggregate relaxation is required. Enforcing the cap $\sum \delta_T + \sum \delta_C \leq V$, Phase 2 produces a feasible minimum-intensity plan. At the optimal V* value of **576.792594** the optimal total beam intensity is **199.6563** with 60/126 beams nonzero. The tumor and critical limits are met after

accounting for the allowable per-pixel slacks, i.e., the instance becomes solvable once minimal deviations are permitted and the plan is then chosen to minimize total radiation.
Takeaway. 2a is feasible on the small dataset and infeasible on the actual dataset; 2b restores feasibility on the actual dataset by allowing the minimal aggregate violation in Phase 1 and then minimizing total intensity under that budget in Phase 2. The resulting plan uses 60 of 126 beamsfor a total intensity of 199.6563.

Task 2(c): Variation of Relaxed Threshold Mathematical Model in AMPL

Objective: Implement your variation of the relaxed threshold mathematical model in AMPL that penalizes radiation delivery to regions close to critical areas. Run the Task 1a model in AMPL on both datasets and report feasibility/results.
AMPL files used.

- `task2c.mod` — model with pixel-level slack variables to allow variation, with their respective penalty levels set to 1, as well as the penalty component for radiation put on the boundary pixels
- `task2c_small.run`, `task2c_actual.run` — two run files that set up the different specs of the different examples and compile the output from the model `2b_small_beam.csv`, `2b_small_intensities.csv`, `2b_actual_beam.csv`, `2b_actual_intensities.csv`.

Small example — feasible.

```
CPLEX 22.1.1:               CPLEX 22.1.1: optimal solution; objective 4.744
4 simplex iteration
```

The small example applied to the model from 1c produces a positive optimal solution of 4.744 for the necessary penalty value for the model to be feasible when accounting for both variation from the required upper and lower radiation limits on non-critical and tumor pixels respectively, as well as when punishing radiation received by non-critical, non-tumorous boundary pixels in the process of treatment. This changes from our previous model where there was an optimal "penalty" value, in the form of the the total value of the imposed slack variables, of 0 since the small example was feasible without having to deviate from the limits and enact a penalty. By including this additional penalty, we see that the optimal "penalty" value rise from 0 since the optimal solution to minimize the total beam intensity is now being affected by the punished radiation to non-critical, non tumor pixels.
Actual example — feasible.

```
CPLEX 22.1.1:               CPLEX 22.1.1: optimal solution; objective 582.6339944
350 simplex iterations
```

The actual example applied to the model from 1c produces a positive optimal solution of 582.6339944 for the necessary penalty value for the model to be feasible when accounting for both variation from the required upper and lower radiation limits on non-critical and tumor pixels respectively, as well as when punishing radiation received by non-critical, non-tumorous boundary pixels in the process of treatment. Given that we already knew that variations from the radiation limit values for the actual example to be feasible, it makes sense that we would see a much higher optimal value for this example when adding an additional penalty component to the problem.